

E-Commerce Sales Trends Analysis Using Microsoft Excel

1. Project Overview

This project analyzes an e-commerce sales dataset containing 34,500 customer orders collected between September 2023 and September 2025. The objective of the project is to evaluate sales performance, customer purchasing behavior, regional contribution, payment preferences, and return trends using Microsoft Excel.

The project demonstrates the use of data cleaning, exploratory data analysis (EDA), Pivot Tables, Pivot Charts, KPI tracking, and dashboard creation to generate meaningful business insights from raw transactional data.

2. Business Problem

E-commerce companies generate large volumes of sales data every day. Without proper analysis, it becomes difficult to identify top-performing products, customer preferences, regional performance, and operational challenges such as product returns.

The objective of this project is to transform raw sales data into actionable business insights that can support data-driven decision-making and business growth.

3. Dataset Information

Dataset Size: 34,500 Records

Number of Columns: 17

Key Attributes

- Order ID
- Customer ID
- Product ID
- Category
- Price
- Discount
- Quantity
- Payment Method
- Order Date
- Delivery Time (Days)
- Region
- Returned
- Total Amount

- Shipping Cost
- Profit Margin
- Customer Age
- Customer Gender

4. Data Cleaning Process

The following data quality checks were performed:

- Checked for missing values.
- Checked for duplicate records.
- Verified data types and date formats.
- Ensured consistency in categorical fields.
- Validated numerical fields for analysis.

No significant data quality issues were identified.

5. Key Performance Indicators (KPIs)

KPI	Value
Total Revenue	₹58,65,293
Total Orders	34,500
Total Quantity Sold	51,430
Average Order Value	₹170
Average Profit Margin	28%
Return Rate	5.52%

6. Analysis Performed

Product Category Analysis

Revenue contribution by category was analyzed to identify the highest-performing product segments.

Monthly Revenue Trend Analysis

Monthly sales trends were analyzed to understand business performance over time.

Regional Performance Analysis

Revenue contribution across regions was evaluated to identify strong and weak markets.

Payment Method Analysis

Customer payment preferences were analyzed using order and revenue data.

Return Analysis

Return patterns were analyzed to understand customer satisfaction and operational performance.

Customer Demographic Analysis

Customer behavior was analyzed based on age groups and gender distribution.

7. Key Insights

Insight 1

Electronics generated ₹33.19 lakh in revenue and contributed 56.59% of total sales, making it the most important revenue-generating category.

Insight 2

The South region generated the highest revenue contribution at 22.13% of total sales.

Insight 3

Credit Card was the most preferred payment method, generating the highest revenue and order volume.

Insight 4

The overall return rate was only 5.52%, indicating strong customer satisfaction and product acceptance.

Insight 5

Fashion recorded the highest number of returned orders, suggesting potential opportunities for quality improvement or customer expectation management.

Insight 6

Customers aged 28–37 contributed the highest share of revenue among all age groups.

Insight 7

Revenue contribution from male and female customers was nearly balanced, indicating a diverse customer base.

Insight 8

Sales performance remained relatively stable throughout the analysis period with consistent monthly revenue generation.

8. Business Recommendations

- Continue investing in Electronics as the primary revenue driver.
- Investigate return patterns within the Fashion category.

- Expand marketing efforts in the South region.
- Promote digital payment methods through targeted offers and incentives.
- Focus customer engagement campaigns on the 28–37 age group.
- Explore opportunities to increase revenue contribution from lower-performing categories.

9. Conclusion

The analysis revealed strong performance in the Electronics category, balanced regional sales distribution, low return rates, and healthy customer diversity. The dashboard and analytical findings provide valuable insights that can support strategic business decisions and improve overall operational performance.

10. Dashboard

